Appendix 1. Database search strategy (August 14th, 2021)

Database	Search
PubMed (n = 2,414)	((“dental implant” [Title/Abstract] OR “dental implants” [MeSH Terms] OR “Implant material” [Title/Abstract] OR “Dental Implant-Abutment Design” [MeSH Terms] OR “Implant material” [Title/Abstract] OR “Implant Surface” [Title/Abstract] OR “dental devices” [Title/Abstract] OR “dental abutments” [MeSH Terms] OR “titanium” [MeSH Terms]) AND (“Biofilms” [MeSH Terms] OR “Biofilm” [Title/Abstract] OR “Dental Deposits” [MeSH Terms] OR “Dental plaque” [MeSH Terms] OR “Plaque” [Title/Abstract] OR “bacterial adhesion” [MeSH Terms] OR “bacterial colonization” [Title/Abstract] OR “oral bacteria” [Title/Abstract] OR “bacteria” [MeSH Terms] OR “bacterial count” [Title/Abstract] OR “Bacterial Load” [MeSH Terms] OR “bacterial attachment” [Title/Abstract] OR “Microbiology” [MeSH Terms] OR “colony count, microbial” [MeSH Terms] OR “microbial” [Title/Abstract] OR “microorganisms” [Title/Abstract] OR “subgingival colonization” [Title/Abstract] OR “initial colonization” [Title/Abstract]) AND (“in vivo” [Title/Abstract] OR “humans” [Title/Abstract] OR “human” [Title/Abstract] OR “patient” [Title/Abstract] OR “patients” [Title/Abstract] OR “volunteers” [Title/Abstract] OR “volunteer” [Title/Abstract])
Scopus (n = 4,032)	(TITLE-ABS-KEY (“dental implant” OR “dental implants” OR “Implant material” OR “Implant material” OR “Implant Surface” OR “dental devices” OR “dental abutments” OR “titanium”) AND TITLE-ABS-KEY (“Biofilms” OR “Biofilm” OR “Dental Deposits” OR “Plaque” OR “bacterial adhesion” OR “bacterial colonization” OR “oral bacteria” OR “bacterial count” OR “Bacterial Load” OR “bacterial attachment” OR “Microbiology” OR “colony count” OR “microbial” OR “microorganisms” OR “subgingival colonization” OR “initial colonization”) AND TITLE-ABS-KEY (“in vivo” OR “humans” OR “human” OR “patient” OR “patients” OR “volunteers” OR “volunteer”) AND NOT TITLE-ABS-KEY (in AND vitro) AND NOT TITLE-ABS-KEY (review))
Web of Science (n = 1,603)	(TS=(“dental implant” OR “dental implants” OR “Implant material” OR “Implant material” OR “Implant Surface” OR “dental devices” OR “dental abutments” OR “titanium”) AND TS=(“Biofilms” OR “Biofilm” OR “Dental Deposits” OR “Plaque” OR “bacterial adhesion” OR “bacterial colonization” OR “oral bacteria” OR “bacterial count” OR “Bacterial Load” OR “bacterial attachment” OR “Microbiology” OR “colony count” OR “microbial” OR “microorganisms” OR “subgingival colonization” OR “initial colonization”) AND TS=(“in vivo” OR “humans” OR “human” OR “patient” OR “patients” OR “volunteers” OR “volunteer”) NOT ALL=(review) NOT ALL=(“in vitro”)) AND (DT==(“ARTICLE”))

Appendix 2. Papers found at electronic data bases and evaluated through full text reading

Source	Included	Reason for exclusion
Conserva et al. [8]	No	Different experimental times of biofilm evaluation
Desch et al. [9]	No	Lack of comparison of smooth machined titanium surface with a rough titanium surface
Gatewood et al. [10]	No	Inclusion of patients with periodontal disease and nicotine consumption
Grössner-Schreiber et al. [11]	No	Lack of comparison of smooth machined titanium surface with a rough titanium surface
Hannig [12]	No	Have not analysed titanium surfaces
Leonhardt et al. [13]	No	Have not analysed titanium surfaces
Macedo et al. [14]	No	Have not analysed titanium surfaces
do Nascimento et al. [15]	No	Lack of comparison of smooth machined titanium surface with a rough titanium surface
do Nascimento et al. [16]	No	Lack of comparison of smooth machined titanium surface with a rough titanium surface
Ferreira-Ribeiro et al. [17]	No	Lack of comparison of smooth machined titanium surface with a rough titanium surface
Rimondini et al. [18]	No	Lack of comparison of smooth machined titanium surface with a rough titanium surface
Scarano et al. [19]	No	Lack of comparison of smooth machined titanium surface with a rough titanium surface
Scotti et al. [20]	No	Evaluation of strategies to destroy previously installed biofilm
Wang et al. [21]	No	Lack of comparison of smooth machined titanium surface with a rough titanium surface
Al-Ahmad et al. [23]	Yes	Not applicable
Al-Ahmad et al. [24]	Yes	Different experimental times of biofilm evaluation
de Freitas et al. [25]	Yes	Not applicable
Herrmann, et al. [28]	Yes	Not applicable

Appendix 3. Papers found through manual searching and evaluated through full text reading

Source	Included	Reason for exclusion
Bürgers et al. [22]	No	Different experimental times of biofilm evaluation
Giordano et al. [26]	Yes	Not applicable
Zaugg et al. [27]	Yes	Not applicable